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An accelerated alternating direction method of multiplier for MRI with TV regularisation

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ABSTRACT

Compressed Sensing (CS) is important in the field of image processing and signal processing, and CS-Magnetic Resonance Imaging (MRI) is used to reconstruct image from undersampled k-space data. Total Variation (TV) regularisation is a common technique to improve the sparsity of image, and the Alternating Direction Multiplier Method (ADMM) plays a key role in the variational image processing problem. This paper aims to improve the quality of MRI and shorten the reconstruction time. We consider MRI to solve a linear inverse problem, we convert it into a constrained optimization problem based on TV regularisation, then an accelerated ADMM is established. Through a series of theoretical derivations, we verify that the algorithm satisfies the convergence rate of $O\left(1/k^2\right)$ under the condition that one objective function is quadratically convex and the other is strongly convex. We select five undersampled templates for testing in MRI experiment and compare it with other algorithms, experimental results show that our proposed method not only improves the running speed but also gives better reconstruction results.

1. Introduction

Image exists as visual media in human life, people can perceive image to obtain information, express ideas and appreciate beauty. Therefore, mature image processing technology will provide convenience for technological progress and human development. However, the image may be degraded in the acquisition, storage and transmission process, due to a variety of factors. For example, image is contaminated by noise, blurred image and reduced resolution. It can be seen that image recovery is crucial, but this process requires solving a linear inverse problem, as follows

$$y = Ms + e,$$

where $M \in R^{m\times n}$ is a linear bounded operator, such as a blur operator or a sampling matrix. $y \in R^m$ is the observed noisy image or data, $s \in R^n$ is the original image, $e \in R^m$ is noise. In order to approximate the original image, scholars at home and abroad have proposed many reconstruction models to minimize e. For example, in the field of digital signal processing, a series of traditional methods in the early days followed the sampling theorem, but as the data to be processed become richer and richer in information and the signal bandwidth becomes wider and wider, the sampling difficulty and complexity increase greatly according to the traditional sampling theory. In 2006, Donoho and Candès [1,2] proposed the theory of CS. In 2007, Lustig et al. [3] proposed sparse MRI, which introduced the CS theory into MRI for the first time, and proposed the CS-based MRI (CS-MRI) technique. A more comprehensive discussion of this technique was provided by Lustig et al. [4] in 2008. Later, with the development of MRI technology, the image quality becames higher and higher, and the application of MRI technology in medical imaging becames more and more widespread.

However, during MRI scanning, the contrast of MR image can be distorted due to the physiological movements of the patient [5]. MRI reconstructs the original image based on sparsity, which is actually seeking the sparsest solution, which is a minimum L_0 paradigm optimization problem, and it is also a NP-Hard problem. Therefore, researchers have introduced regularisation techniques such as L_1 norm, L_2 norm and total variation to achieve MRI. A commonly used total variation minimization model [6,7] is as follows

$$min\frac{1}{2}\|Ms - y\|_2^2 + \mu\|s\|_{TV} + \frac{\varepsilon}{2}\|\nabla s\|_2^2, \tag{1}$$

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0730-725X/© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

where $\|s\|_{TV} = \|\nabla s\|_1$, $\mu, \varepsilon > 0$. ∇ is finite difference operator. If s is a vector, then $(\nabla s)_i = (s_{i+1} - s_i)$. If s is an image, then $(\nabla s)_{i,j} = \left(s_{i+1,j} - s_{i,j}, s_{i,j+1} - s_{i,j}\right)$. In addition, since MR image satisfy sparsity, a sparser representation can be sought under the transformation of sparse basis. The paper [8] proposed a sparsified representation of block-based directional wavelets, and paper [9] weighted the coefficients of conventional sampling wavelets to obtain a sparse representation of MR image. In order to better reconstruct the texture and edges of MR image, Jiang et al. [10] used image block-based curvilinear wavelet transform to sparse the MR image. Riaz et al. [11] proposed the oblique wavelet transform to sparse the image, and compared with the traditional wavelet transform and discrete cosine transform to achieve better reconstruction results.

Reconstruction algorithms are also important in the study of CS-MRI, and most of the optimization algorithms can be applied to MRI. The iterative hard-thresholding method [12], as a well-known MRI algorithm, although its algorithm is simple, the reconstruction results are not satisfactory enough. Zhang et al. [13] established a Conjugate Gradient (CG) method for CS-MRI. Mousavi et al. [14] designed two types of three conjugate gradient methods for medical imaging. More CG methods for image restoration or MRI, see [15,16,17]. CG methods is used in MRI, which can obtain better reconstruction results, but the convergence speed is slow and the iteration step size depends on the line search. The Iterative Shrinkage Thresholding Algorithm (ISTA) is a widely used iterative algorithm, but its convergence speed is slow. The Fast Iterative Shrinkage Thresholding Algorithm (FISTA) [18] and its improved algorithms are more widely used [19,20], but these algorithms and their accelerated forms are suitable for solving the CS-MRI problem with a single regularisation term, and if there are sparse models with more than one regularisation term, they need to be solved by parallel computation or alternating miniaturisation. Goldstein and Osher [21] proposed a split Bregman algorithm, and applied it to image denoising and MRI. Xiaoyu et al. [22] established a new MRI model, and solved it with inertial neighbourhood algorithm. Ramani and Fessler [23] proposed the augmented Lagrangian algorithm, which is also used in MRI. Hong et al. [24] proposed a quasi-newton proximal method on complex spaces, and verified the effectiveness on reconstructing non-Cartesian MRI. The ADMM [25,26] is based on the augmented Lagrange multiplier method, which decomposes the original problem into easy-to-solve minimization subproblems, and then solves them by using the alternating minimization method, which has faster convergence speed, and also has good applications in CS-MRI [27,28,29,30].

In this paper, we consider model (1) to solve the TV regularised MRI problem, but it is difficult to solve it directly, so we transform it into the form of a constrained optimization problem

$$\begin{aligned} &min\frac{1}{2}\parallel Ms - y\|_2^2 + \mu\parallel t\|_1 + \frac{\varepsilon}{2}\parallel t\|_2^2, \\ &\text{s.t.}\nabla s - t = 0. \end{aligned} \tag{2}$$

It can be shown that (2) satisfies the form of the separable convex minimization problem

$$\begin{aligned} &minF(s) + G(t), \\ &s.t.\ \ \text{As} + Bt = b, \end{aligned} \tag{3}$$

where

$$\begin{aligned} &F(s) = \frac{1}{2}\|Ms - y\|_2^2, G(t) = \mu\|t\|_1 + \frac{\varepsilon}{2}\|t\|_2^2, \\ &A = \nabla, B = -I, b = 0. \end{aligned} \tag{4}$$

It is clear that $F(s)$ is quadratic convex function, $G(t)$ is strongly convex function, $A \in R^{l\times n}$, $B \in R^{l\times l}$, $s \in R^n$, $t \in R^l$, $b \in R^l$.

Since objective function of problem (3) consists of two parts, the dual decomposition method and the augmented Lagrange multiplier method [31] have been combined to propose ADMM, which has good convergence and convergence speed, and the iterative formula of the classical ADMM is as follows

$$\begin{aligned} &s_{k+1} = \underset{s\in R^n}{\text{argmin}}\left\{F(s) + \langle\lambda_k, -As\rangle + \frac{\beta}{2}\parallel b - As - Bt_k\|^2\right\}, \\ &t_{k+1} = \underset{t\in R^l}{\text{argmin}}\left\{G(t) + \langle\lambda_k, -Bt\rangle + \frac{\beta}{2}\parallel b - As_{k+1} - Bt\|^2\right\}, \\ &\lambda_{k+1} = \lambda_k + \beta(b - As_{k+1} - Bt_{k+1}). \end{aligned}$$

ADMM firstly decomposes subproblems and then combines the solutions. Many scholars have proposed a series of improved acceleration algorithms to increase the running speed of ADMM. For example, a new fast algorithm based on variable splitting combining ALM and ADMM is proposed - Split Augmented Lagrangian Shrinkage Algorithm (SALSA) [32], which is used in MRI and image denoising. Bartz et al. [33] paid attention to the step-size defect of ADMM, considering the case of one objective function is strongly convex and the other is weakly convex, then proposed an adaptive ADMM (aADMM) with adaptive step-size, which can flexibly adjust the range of the penalty parameter, thus making the algorithm more adaptable. In addition, an accelerated symmetric ADMM for solving nonconvex nonsmooth problems was proposed in [34] and applied to signal processing. In recent years, Ding et al. [35] differed from usual as they worked on solving the dyadic model of the original model, combined with the symmetric Gauss-Seidel (sGS) technique, to propose a variant of ADMM and apply it to MRI. Yang et al. [36] proposed two novel deep networks based on ADMM for MRI model, which are effective on MRI experiments, while Yiasemis et al. [37] combined ADMM with neural networks to achieve high reconstruction fidelity in cardiac MRI experiments. In addition, a Plug-and-Play ADMM (PnP-ADMM) was proposed in [38] for MRI reconstruction with Convex Non-Convex (CNC) sparse regularisation, while a stage-residual PnP-ADMM based on truncated residuals was proposed in [39].

Nesterov acceleration technique [40] is often used in optimization algorithms. Such as the FISTA algorithm [18] is the first algorithm to incorporate Nesterov acceleration technique. Goldstein et al. [41] proposed fast ADMM with Nesterov acceleration technique under the condition that both objective functions are strongly convex, and compared it with classical ADMM [26] and Alternating Minimization Algorithm (AMA) [42], which can achieve better image processing results.

Meng and Zhang [43] used the conjugate function to weaken the requirement of convexity of the objective function in [41]. It only requires one objective function to be strongly convex and the other is linear, and also weakens the penalty parameter setting, obtaining a better convergence speed. But it appended additional information about the iteration points of classical ADMM to complete the convergence proofs.

These problems are completed in this paper. By combining the ideas of [41] and [43] and introducing Nesterov acceleration technique, based on TV regularisation model (1), we propose a New Accelerated ADMM (NAADMM) to solve special MRI problem (3). We obtain a convergence result of $O\left(1/k^2\right)$ under the condition that one objective function is strongly convex and the other is quadratic convex.

The paper is organized as follows: the preliminary knowledge and the NAADMM algorithm for MRI with TV regularisation in Section 2. Section 3 gives the convergence analysis of the NAADMM algorithm. Section 4 performs MRI experiment to show our algorithm is meaningful. Section 5 gives the conclusion of this paper.

2. Material and methods

2.1. Preliminaries

In this section, we give some preliminary knowledge that will be needed for this paper.

Definition 2.1. [44] (Moore–Penrose Pseudoinverse) Let $A \in C^{m\times n}$, if $G \in C^{n\times m}$ satisfies

$$AGA = A, \tag{5}$$

$$GAG = G, \tag{6}$$

$$(AG)^H = AG, \tag{7}$$

$$(GA)^H = GA, \tag{8}$$

then G is called the M-P generalized inverse of A, denoted A^+ or $A^\dagger$, and A^+ is unique. The above four equations are called the M-P equations.

Theorem 2.1. [44] (Matrix Singular Value Decomposition) Let the singular value decomposition of $A \in C_r^{m\times n}$ (be)

$$A = U\begin{pmatrix} S_r & 0 \\ 0 & 0 \end{pmatrix}V^H,$$

where $U \in C^{m\times m}, V \in C^{n\times n}$ are unitary matrix, $S_r = diag(\sigma_1, \sigma_2, \ldots, \sigma_r)$, $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > 0$ are the positive singular value, and r is the rank of A. Then we have $A^+ = V\begin{pmatrix} S_r^{-1} & 0 \\ 0 & 0 \end{pmatrix}U^H, \|A^+\|_2 = \frac{1}{min_{1\leq i\leq r}\{\sigma_i\}}$.

Definition 2.2. [48] A differentiable function $f : C \subseteq R^n \to R$ is said to have Lipschitz continuous gradient if there exists $L_{\nabla f} > 0$, such that

$$\|\nabla f(x) - \nabla f(y)\| \leq L_{\nabla f}\|x - y\|, \forall x, y \in C,$$

where C is the domain of the function f. A function is said to be $L_{\nabla f}-$ smooth if it has Lipschitz continuous gradient with constant $L_{\nabla f}$.

Definition 2.3. [48] A function $f : C \subseteq R^n \to R$ is called strongly convex if C is a convex set and there exists a constant $\mu > 0$, for any $x, y \in C$ and $\alpha \in [0, 1]$, we have

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y) - \frac{\mu\alpha(1-\alpha)}{2}\|x - y\|^2,$$

where μ is called the strong convexity factor of f.

Definition 2.4. [47] Given $f : C \subseteq R^n \to R$, its conjugate function is defined as

$$f^*(y) = \sup_{x\in C}\{\langle y, x\rangle - f(x)\}.$$

If f is a closed convex function, then $f^{**} = f$. Note that the function f^* is always a closed convex function, whether or not f is convex.

Proposition 2.1. [47] (Differential Properties Of Conjugate Functions) If a function f is differentiable, then f^* is also differentiable, and for $\forall x, y \in R^n$, with $y = \nabla f(x)$, there are $x = \nabla f^*(y)$.

Definition 2.5. [51] (Affine Function) If a function $f : C \subseteq R^n \to R$ is affine, then for any $x_i \in R^n, \theta_i \in R, i = 1, 2, 3\ldots k$, and $\sum_{i=1}^k \theta_i = 1$, there exists

$$f\left(\sum_{i=1}^k \theta_i x_i\right) = \sum_{i=1}^k \theta_i f(x_i).$$

Lemma 2.1. [45] *Let a function $f : \chi \to (-\infty, +\infty]$ be defined on a non-empty convex set $\chi \subseteq R^n$, if f is continuously differentiable, for any $(x, y) \in \chi \times \chi$, then f is a convex function if and only if one of the following conditions is satisfied.*

a. *first-order condition*

$$f(y) \geq f(x) + \langle \nabla f(x), y - x\rangle.$$

b. *second-order condition*

$$\langle \nabla^2 f(x)(y - x), y - x\rangle \geq 0.$$

Lemma 2.2. [49] *If $f : C \subseteq R^n \to R$ is an $L_{\nabla f}-$ smooth function, then there are*

$$|f(y) - f(x) - \langle \nabla f(x), y - x\rangle| \leq \frac{L_{\nabla f}}{2}\|x - y\|^2, \forall x, y \in C.$$

Lemma 2.3. [47] *If $f : C \to R$ is a μ_f- strongly convex function, then*

$$f(y) \geq f(x) + \langle g, y - x\rangle + \frac{\mu_f}{2}\|y - x\|^2, \forall g \in \partial f(x).$$

In particular, if f is μ_f- strongly convex function, $x^* = argmin_x f(x)$, then

$$f(x) - f(x^*) \geq \frac{\mu_f}{2}\|x - x^*\|^2.$$

Lemma 2.4. [41] *If a function f is strongly convex, then its conjugate function has Lipschitz continuous gradient*

$$\|\nabla f^*(x) - \nabla f^*(y)\| \leq L_{\nabla f^*}\|x - y\|,$$

where $L_{\nabla f^*}$ denotes the Lipschitz constant. More properties of strongly convex functions can be found in [46].

Definition 2.6. [50] For a convex function h, its proximal operator is

$$prox_h(x) = \underset{u\in domh}{argmin}\left\{h(u) + \frac{1}{2}\|u - x\|^2\right\}.$$

The purpose of the proximal operator is to solve for a point that is not too far away from x, while keeping the value of the function $h(x)$ small. The definition of the proximal operator makes sense if the solution to the optimization problem in this definition is unique.

Lemma 2.5. [50] (*Relationship Between Proximal Operator and Subgradient*) *If h is a proper closed convex function, then*

$$u = prox_h(x) \Leftrightarrow x - u \in \partial h(u),$$

where $\partial h(u)$ is the subgradient of $h(x)$ at u.

Example 2.1. [50] (Example Of Proximal Operator) If $h(x) = t\|x\|_1$, the constant $t > 0$ is a positive real number, then there are

$$prox_h(x) = sign(x)max\{|x| - t, 0\} = shrink(x, t),$$

where *shrink* is called the soft threshold operator.

Lemma 2.6. [50] (Property of conjugate function of strongly convex function) Let $f(x)$ is proper and closed strongly convex function, $f^*(y)$ is conjugate function of $f(x)$. Then $f^*(y)$ is differentiable function and the gradient is the corresponding maximun point x.

2.2. Accelerated ADMM for MRI with TV regularisation

In this section, we solve MRI problem (3) using the following NAADMM algorithm proposed in this paper.

Algorithm 1. NAADMM.

(NAADMM)
Require:
$\lambda_0 = \hat{\lambda}_1 \in R^l, t_0 = \hat{t}_1 = \nabla G^*\left(B^T\lambda_0\right) \in R^l,$
$1 - \sqrt{\frac{1}{6}} < \frac{\eta}{\beta} < 1, \alpha_1 = 1.$
1: **for** $k = 1, 2, 3 \cdots$ **do.**

$$2: \ s_k = \underset{s\in R^n}{argmin}\left\{F(s) + \left\langle \hat{\lambda}_k, -As\right\rangle + \frac{\beta}{2}\parallel b - As - B\hat{t}_k\parallel^2\right\}. \tag{9}$$

$$3: \ t_k = \underset{t\in R^l}{argmin}\left\{G(t) + \left\langle \hat{\lambda}_k, -Bt\right\rangle + \frac{\beta}{2}\parallel b - As_k - Bt\parallel^2\right\}. \tag{10}$$

$$4: \ \lambda_k = \hat{\lambda}_k + \beta(b - As_k - Bt_k). \tag{11}$$

$$5: \ \alpha_{k+1} = \frac{1 + \sqrt{1 + 4\alpha_k^2}}{2}. \tag{12}$$

$$6: \ \hat{t}_{k+1} = t_k + \frac{\alpha_k - 1}{\alpha_{k+1}}(t_k - t_{k-1}) + \frac{\alpha_k}{\alpha_{k+1}}(1 - \frac{\eta}{\beta})(\hat{t}_k - t_k). \tag{13}$$

$$7: \ \hat{\lambda}_{k+1} = \lambda_k + \frac{\alpha_k - 1}{\alpha_{k+1}}(\lambda_k - \lambda_{k-1}) + \frac{\alpha_k}{\alpha_{k+1}}(1 - \frac{\eta}{\beta})(\hat{\lambda}_k - \lambda_k). \tag{14}$$

8: **end for.**

Note: For problem (3) studied in this paper, we will give explicit solutions to subproblems (9) and (10), where $M = R\mathscr{F}$, s stands for MR brain image, R stands for the undersampling operator, $\mathscr{F}$ is the Fourier operator and y is the undersampled Fourier measure.

For (9) in Algorithm 1, there are

$$\begin{aligned} s_k &= \underset{s\in R^n}{argmin}\left\{F(s) + \langle\widehat{\lambda}_k, -As\rangle + \frac{\beta}{2}\parallel b - As - B\widehat{t}_k\parallel^2\right\} \\ &= \underset{s\in R^n}{argmin}\left\{\frac{1}{2}\parallel R\mathscr{F}s - y\parallel_2^2 - \widehat{\lambda}_k^T\nabla s + \frac{\beta}{2}\parallel \widehat{t}_k - \nabla s\parallel^2\right\} \\ &= \underset{s\in R^n}{argmin}\left\{\frac{1}{2}\left((R\mathscr{F}s - y)^T(R\mathscr{F}s - y) - \widehat{\lambda}_k^T\nabla s + \frac{\beta}{2}\left((\widehat{t}_k - \nabla s)^T(\widehat{t}_k - \nabla s)\right\}.\right.\right. \end{aligned}$$

Since $F(s)$ is smooth and differentiable, using the optimality condition in the above equation, we obtain

$$0 = (R\mathscr{F}s_k)^T R\mathscr{F} - y^T R\mathscr{F} - \nabla^T\widehat{\lambda}_k + \beta\nabla^T\nabla s_k - \beta\nabla^T\widehat{t}_k.$$

Since $\mathscr{F}^T = \mathscr{F}^{-1}$, there are

$$\begin{aligned} &\left[(R\mathscr{F})^T R\mathscr{F} + \beta\nabla^T\nabla\right]s_k = y^T R\mathscr{F} + \nabla^T\widehat{\lambda}_k + \beta\nabla^T\widehat{t}_k \\ &\mathscr{F}^T\left(R^T R + \beta\nabla^T\nabla\right)\mathscr{F}s_k = y^T R\mathscr{F} + \nabla^T\widehat{\lambda}_k + \beta\nabla^T\widehat{t}_k. \end{aligned}$$

To simplify, let $Q = R^T R + \beta\nabla^T\nabla$, $q = y^T R\mathscr{F} + \nabla^T\widehat{\lambda}_k + \beta\nabla^T\widehat{t}_k$, i.e.,

$$\mathscr{F}^{-1}Q\mathscr{F}s_k = q.$$

Thus have

$$s_k = \mathscr{F}^{-1}Q^{-1}\mathscr{F}q. \tag{15}$$

For (10) in Algorithm 1, there are

$$\begin{aligned} t_k &= \underset{t\in R^l}{argmin}\left\{G(t) + \langle\widehat{\lambda}_k, -Bt\rangle + \frac{\beta}{2}\parallel b - As_k - Bt\parallel^2\right\} \\ &= \underset{t\in R^l}{argmin}\left\{\mu\parallel t\parallel_1 + \frac{\varepsilon}{2}\parallel t\parallel_2^2 + \widehat{\lambda}_k^T t + \frac{\beta}{2}\parallel t - \nabla s_k\parallel^2\right\} \\ &= \underset{t\in R^l}{argmin}\left\{\mu\parallel t\parallel_1 + \frac{\varepsilon}{2}\parallel t\parallel_2^2 + \widehat{\lambda}_k^T t + \frac{\beta}{2}\parallel \nabla s_k\parallel_2^2 + \frac{\beta}{2}\parallel t\parallel_2^2 - \beta t^T\nabla s_k\right\}. \end{aligned}$$

Since $\|\cdot\|_1$ is non-smooth, we consider the soft threshold function in Example 2.1, which makes

$$\begin{aligned} \mathrm{t}_k &= \underset{t\in R^l}{argmin}\left\{\frac{\mu}{\beta+\varepsilon}\parallel t\parallel_1 + \frac{1}{2}\parallel t - \left[\frac{\beta}{\beta+\varepsilon}\left(\nabla s_k - \frac{1}{\beta}\widehat{\lambda}_k\right)\right]\parallel^2\right\} \\ &= shrink\left[\frac{\beta}{\beta+\varepsilon}\left(\nabla s_k - \frac{1}{\beta}\widehat{\lambda}_k\right), \frac{\mu}{\beta+\varepsilon}\right]. \end{aligned}$$

Clearly, (10) in Algorithm 1 simplifies as follows

$$t_k = shrink\left[\frac{\beta}{\beta+\varepsilon}\left(\nabla s_k - \frac{1}{\beta}\widehat{\lambda}_k\right), \frac{\mu}{\beta+\varepsilon}\right]. \tag{16}$$

Therefore, NAADMM algorithm is easy to solve MRI problem with TV regularisation.

3. Results

In the following, we will discuss the convergence properties of Algorithm 1. Since problem (3) solved in this paper, R stands for the undersampling operator and $\mathscr{F}$ is the Fourier operator. Since undersampling operator R is usually used to reduce the resolution of an image, it leads to loss of information, which means that $R^T R$ may have some zero eigenvalues and thus $M^T\mathrm{M}$ is not full rank. Therefore, in the later proof, we need to consider $\left(M^T\mathrm{M}\right)^+$.

Lemma 3.1. *For* $\forall\left(M^T\mathrm{M}\right) \in \mathrm{C}^{n\times n}$, *then the M-P generalized inverse of* $M^T\mathrm{M}$ *exists and is unique,*

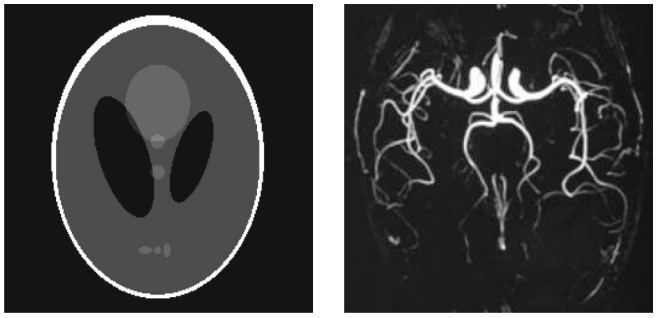

Fig. 1. Original images of Shepp Logan and Brain Angiograph.

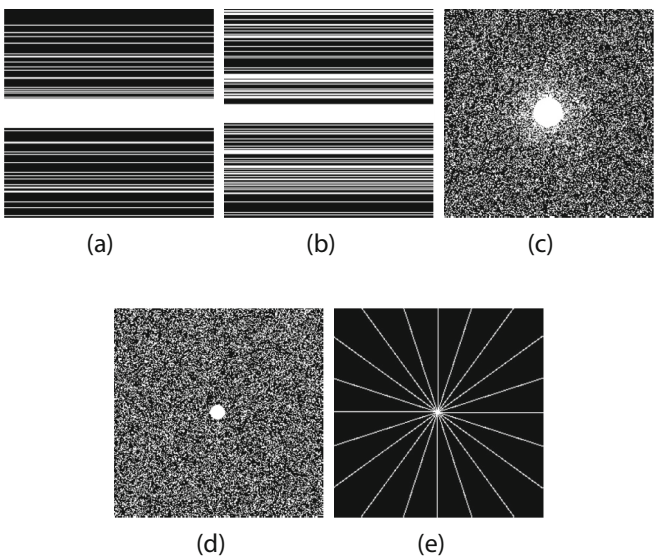

Fig. 2. (a)-(e) show the five undersampling templates (i)-(v), respectively.

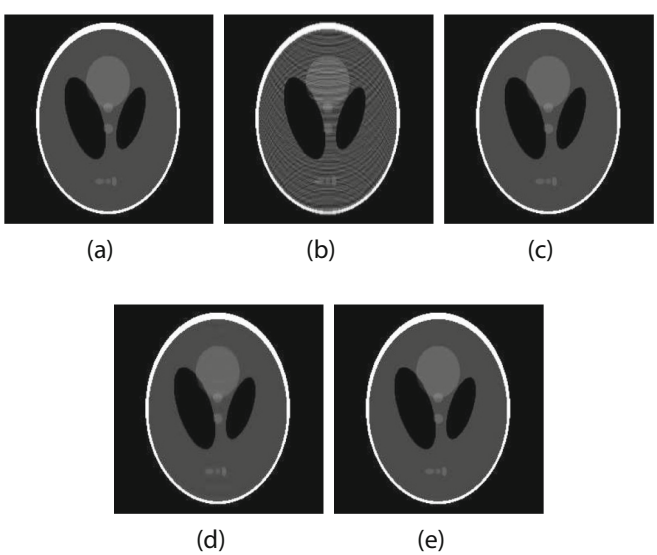

Fig. 3. (a)-(e) show the reconstruction images of the ADMM, AMA, FISTA, aADMM and NAADMM algorithms for (i), respectively.

$$(M^T\mathrm{M})^+ = V\begin{pmatrix} S_r^{-1} & 0 \\ 0 & 0 \end{pmatrix}U^H,$$

where $U \in C^{n\times n}, V \in C^{n\times n}$ are unitary matrix, $S_r = diag(\sigma_1, \sigma_2, \ldots, \sigma_r)$, $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > 0$, r is the number of positive singular values of $M^T\mathrm{M}$.

Proof: By Theorem 2.1, $M^T\mathrm{M}$ has the following singular value decomposition

$$M^T\mathrm{M} = U\begin{pmatrix} S_r & 0 \\ 0 & 0 \end{pmatrix}V^H,$$

it is obvious that $(M^T\mathrm{M})^+ = V\begin{pmatrix} S_r^{-1} & 0 \\ 0 & 0 \end{pmatrix}U^H$ satisfies (5)–(8). Moreover, by Definition 2.1, we can get that $(M^T\mathrm{M})^+$ is unique.

In conclusion, $(M^T\mathrm{M})^+$ exists and is unique.

The proof has been completed. □

Since the objective functions F and G are special functions, we can obtain that their conjugate functions satisfy the following properties.

Lemma 3.2. For problem (3), F^* and G^* have Lipschitz continuous gradients with constants $L_{\nabla F^*} = \left\|(M^TM)^+\right\|_2 = \frac{1}{min_{1\leq i\leq r}\{\sigma_i\}}$ and $L_{\nabla G^*} = \frac{1}{\varepsilon}$, where σ_i is positive singular value of $M^T\mathrm{M}$.

Proof: See Appendix A. □

Before giving the convergence analysis of Algorithm 1, we give the following notations for convenience.

First, we rely on dual theory to give dual problem of (3) according to Definition 2.4

$$\begin{aligned}
&\max_{\lambda\in R^n}\ \min_{s\in R^n, t\in R^l}\{F(s) + G(t) + \langle\lambda, b - As - Bt\rangle\} \\
&= \max_{\lambda\in R^l} -\left\{\max_{s\in R^n, t\in R^l}\langle\lambda, As\rangle + \langle\lambda, Bt\rangle - \langle\lambda, b\rangle - F(s) - G(t)\right\} \\
&= \max_{\lambda\in R^l}\left\{-F^*\left(A^T\lambda\right) - G^*\left(B^T\lambda\right) + \langle\lambda, b\rangle\right\} \\
&= \max_{\lambda\in R^l}\{D(\lambda)\},
\end{aligned}$$

where

$$D(\lambda) = -F^*\left(A^T\lambda\right) - G^*\left(B^T\lambda\right) + \langle\lambda, b\rangle. \tag{17}$$

Since the solution of optimization problem can be described by the KKT condition [40], we give the KKT condition of problem (3).

$$b - As^* - Bt^* = 0, \tag{18}$$

$$\nabla F(s^*) - A^T\lambda^* = 0, \tag{19}$$

$$\nabla G(t^*) - B^T\lambda^* = 0, \tag{20}$$

where (s^*, t^*) is the KKT point of problem (3) and λ^* is the KKT multiplier.

Theorem 3.1. *For* (17), *the following conclusion holds*

$$\nabla D(\lambda^*) = 0,$$

where λ^* is the KKT multiplier of original problem (3).

Proof: By (17), we have

$$D(\lambda^*) = -F^*\left(A^T\lambda^*\right) - G^*\left(B^T\lambda^*\right) + \langle\lambda^*, b\rangle.$$

Therefore

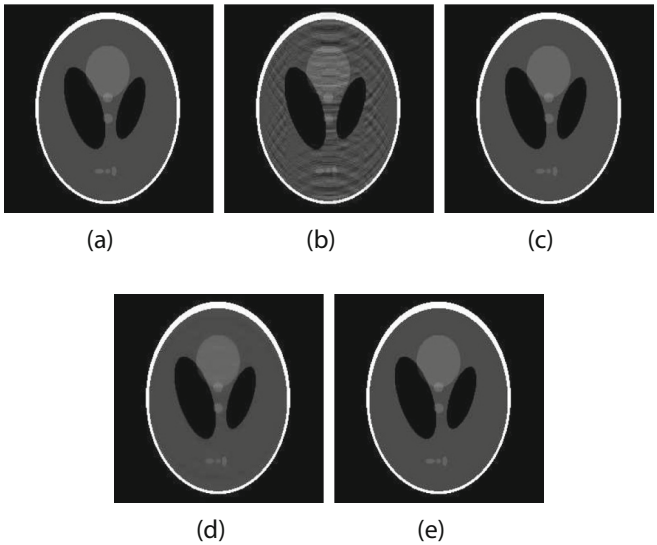

Fig. 4. (a)-(e) show the reconstruction images of the ADMM, AMA, FISTA, aADMM and NAADMM algorithms for (ii), respectively.

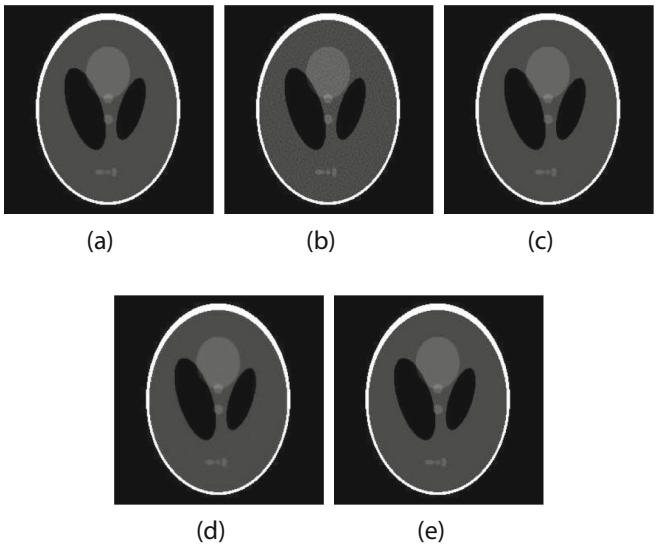

Fig. 5. (a)-(e) show the reconstruction images of the ADMM, AMA, FISTA, aADMM and NAADMM algorithms for (iii), respectively.

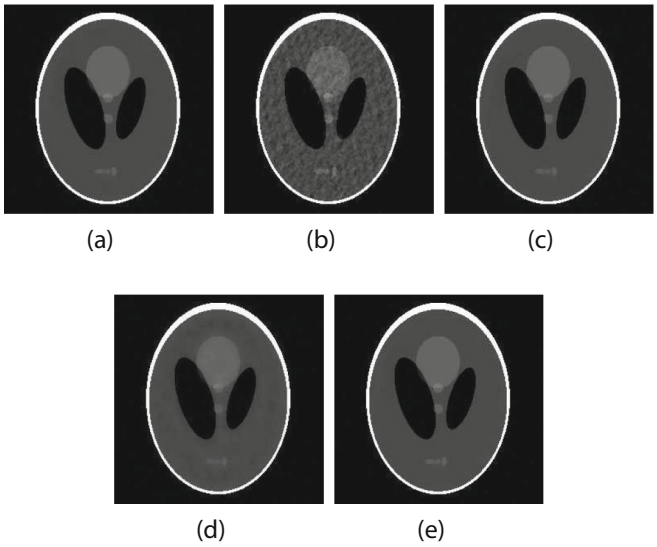

Fig. 6. (a)-(e) show the reconstruction images of the ADMM, AMA, FISTA, aADMM and NAADMM algorithms for (iv), respectively.

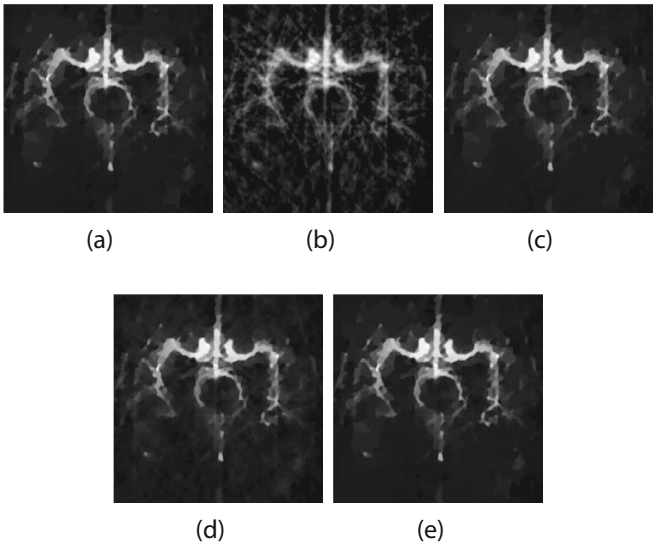

Fig. 7. (a)-(e) show the reconstruction images of the ADMM, AMA, FISTA, aADMM and NAADMM algorithms for (v), respectively.

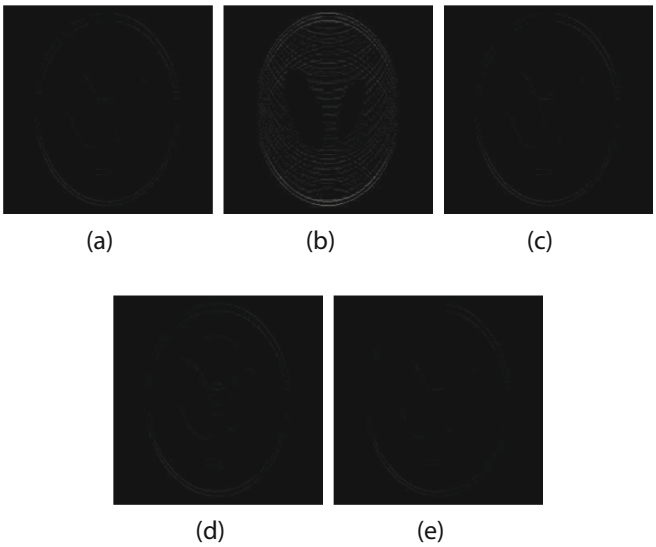

Fig. 8. (a)-(e) show the error images of the ADMM, AMA, FISTA, aADMM and NAADMM algorithms for (i), respectively.

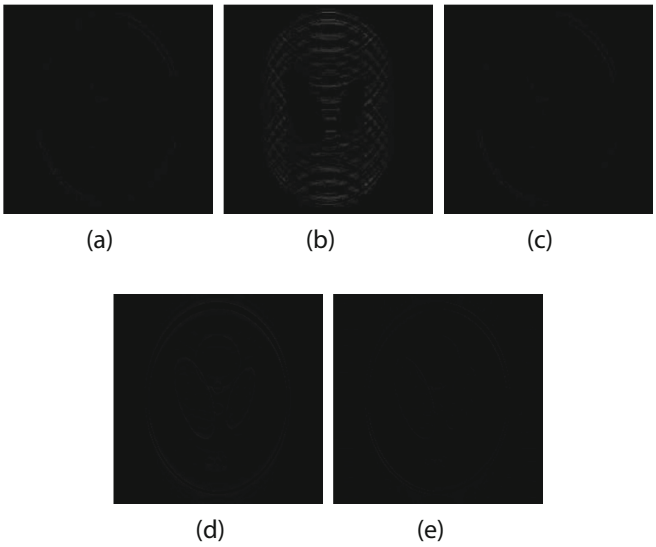

Fig. 9. (a)-(e) show the error images of the ADMM, AMA, FISTA, aADMM and NAADMM algorithms for (ii), respectively.

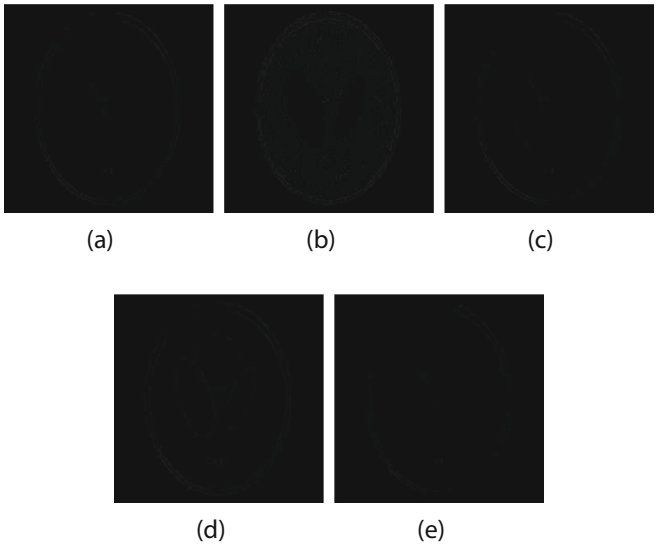

Fig. 10. (a)-(e) show the error images of the ADMM, AMA, FISTA, aADMM and NAADMM algorithms for (iii), respectively.

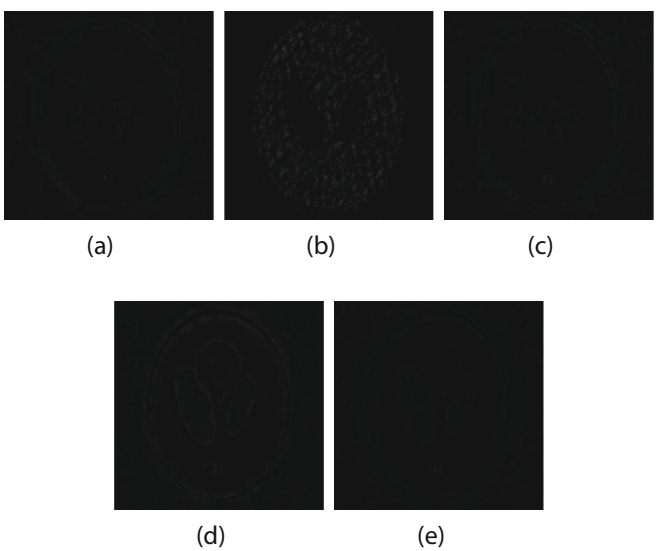

Fig. 11. (a)-(e) show the error images of the ADMM, AMA, FISTA, aADMM and NAADMM algorithms for (iv), respectively.

$$\nabla D(\lambda^*) = -A\nabla F^*\left(A^T\lambda^*\right) - B\nabla G^*\left(B^T\lambda^*\right) + b. \tag{21}$$

Combining Proposition 2.1, (19) and (20), it is clear that

$$s^* = \nabla F^*\left(A^T\lambda^*\right),$$
$$t^* = \nabla G^*\left(B^T\lambda^*\right).$$

Substituting above equations into (21), i.e.,

$$\nabla D(\lambda^*) = -As^* - Bt^* + b.$$

Substituting (18) into the above equation, we can get

$$\nabla D(\lambda^*) = 0.$$

The proof has been completed. □

In the following we will give a series of lemmas in order to obtain convergence results for Algorithm 1.

Lemma 3.3. *$\forall t_k, \widehat{t}_k \in R^l, \widehat{\lambda}_k \in R^l, s_k \in R^n$, defining $\overline{\lambda} = \widehat{\lambda}_k + \beta(b - As_k - B\widehat{t}_k)$, the following conclusions hold*

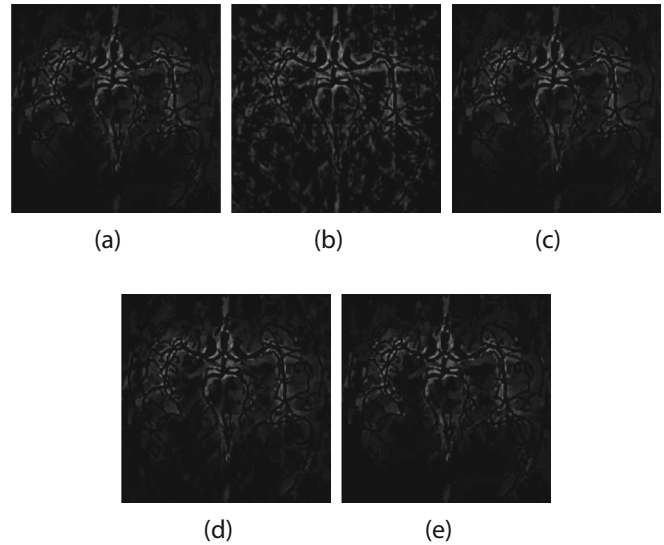

Fig. 12. (a)-(e) show the error images of the ADMM, AMA, FISTA, aADMM and NAADMM algorithms for (v), respectively.

$$s_k = \nabla F^*\left(A^T\overline{\lambda}\right), \tag{22}$$

$$t_k = \nabla G^*\left(B^T\lambda_k\right), \tag{23}$$

$$\widehat{t}_k = \nabla G^*\left(B^T\widehat{\lambda}_k\right). \tag{24}$$

Proof: See Appendix B. □

Lemma 3.4. *If $0 < \beta^3 \le \frac{\varepsilon^2}{L_{\nabla F^*}\|A\|^2}$, for $\forall \kappa \in R^l$, Algorithm 1 produces iterative sequences $\{\lambda_k\}_{k=0}^{\infty}$ and $\{\widehat{\lambda}_k\}_{k=0}^{\infty}$, which satisfy that*

$$D(\lambda_k) - D(\kappa) \ge \frac{1}{\beta}\langle\widehat{\lambda}_k - \lambda_k, \kappa - \widehat{\lambda}_k\rangle + \frac{1}{2\beta}\|\widehat{\lambda}_k - \lambda_k\|^2, \tag{25}$$

where $D(\lambda)$ is defined by (17), $L_{\nabla F^*}$ is given by Lemma 3.2.

Proof: See Appendix C. □

Lemma 3.5. *The iterative sequence $\{(\alpha_k, \lambda_k, \widehat{\lambda}_k)\}_{k=0}^{\infty}$ produced by Algorithm 1 satisfies the following conclusion*

$$\| m_{k+1}\|^2 + 2\eta\alpha_{k+1}^2[D(\lambda^*) - D(\lambda_{k+1})] \le \| m_k\|^2 + 2\eta\alpha_k^2[D(\lambda^*) - D(\lambda_k)],$$

where $m_k = \frac{\eta\alpha_k}{\beta}(\lambda_k - \widehat{\lambda}_k) + (\widehat{\lambda}_k - \widehat{\lambda}_{k-1})(\alpha_k - 1) + \widehat{\lambda}_k - \lambda^*$, $\eta \in R/\{0\}$ and $0 < \eta < \beta$.

Proof : See Appendix D. □

In the following, we will discuss the sequence $\{\lambda_k\}_{k=0}^{\infty}$ produced by Algorithm 1, which makes the objective function of dual problem of original problem 3 is convergent.

Theorem 3.2. *The iterative sequence $\{\lambda_k\}_{k=0}^{\infty}$ produced by Algorithm 1, which satisfy that*

$$D(\lambda^*) - D(\lambda_{k+1}) \le \frac{\frac{4\eta}{\beta^2}\|\lambda_1 - \widehat{\lambda}_1\|^2 + \frac{2\eta+4\beta}{\eta\beta}\|\widehat{\lambda}_1 - \lambda^*\|^2}{(k+2)^2}.$$

Proof : See Appendix E. □

Next we consider convergence from original problem (3). We would like Algorithm 1 to satisfy the primal feasibility condition (18), the dual feasibility condition (19) and (20) as much as possible. However, substituting (11) from Algorithm 1 into the optimality condition for s_k in (9),we can obtain

Table 1
Numerical results of MRI experiments.

Algorithm	$Tol = 1e^{-4}$			$Tol = 1e^{-5}$		
	CPU	*Error*	*PSNR*	*CPU*	*Error*	*PSNR*
***Shepp Logan*(*i*)**						
ADMM	4.33	0.0431	39.4630	7.45	0.0384	40.4902
AMA	6.18	0.1663	27.7521	7.01	0.1661	27.7602
FISTA	5.43	0.0436	39.3764	6.03	0.0436	39.3764
aADMM	5.69	0.0423	39.6443	6.03	0.0423	39.6443
NAADMM	**2.15**	**0.0346**	**41.3631**	**4.08**	**0.0376**	**40.6797**
***Shepp Logan*(*ii*)**						
ADMM	2.53	0.0259	43.8432	8.23	0.0222	45.2486
AMA	6.44	0.1171	30.7968	7.55	0.1171	30.7968
FISTA	5.48	0.0248	44.2811	6.17	0.0248	44.2811
aADMM	4.96	0.0302	42.5800	5.97	0.0302	42.5802
NAADMM	**1.66**	**0.0210**	**45.6997**	**4.33**	**0.0221**	**45.2813**
***Shepp Logan*(*iii*)**						
ADMM	3.06	0.0326	41.8426	6.02	0.0273	43.4414
AMA	4.02	0.0700	35.2355	7.22	0.0549	37.3632
FISTA	5.99	0.0289	42.9510	6.02	0.0289	42.9510
aADMM	5.61	0.0311	42.3039	6.31	0.0311	42.3105
NAADMM	**1.74**	**0.0256**	**43.9726**	**4.21**	**0.0270**	**43.5533**
***Shepp Logan*(*iv*)**						
ADMM	2.75	0.0433	39.4018	6.62	0.0390	40.3577
AMA	7.20	0.0948	32.6122	7.62	0.0948	32.6122
FISTA	6.16	0.0444	39.2316	6.23	0.0444	39.2316
aADMM	4.56	0.0565	37.1358	6.22	0.0566	37.1232
NAADMM	**1.60**	**0.0372**	**40.7477**	**4.31**	**0.0387**	**40.4138**
***Brain Angiograph*(*v*)**						
ADMM	4.64	0.3954	22.1129	7.86	0.3940	22.1443
AMA	**1.81**	0.4632	20.7385	5.61	0.4628	20.7456
FISTA	5.81	0.4048	21.9080	6.10	0.4048	21.9080
aADMM	6.17	0.4088	21.8239	6.11	0.4088	21.8239
NAADMM	3.14	**0.3931**	**22.1635**	**4.19**	**0.3932**	**22.1617**

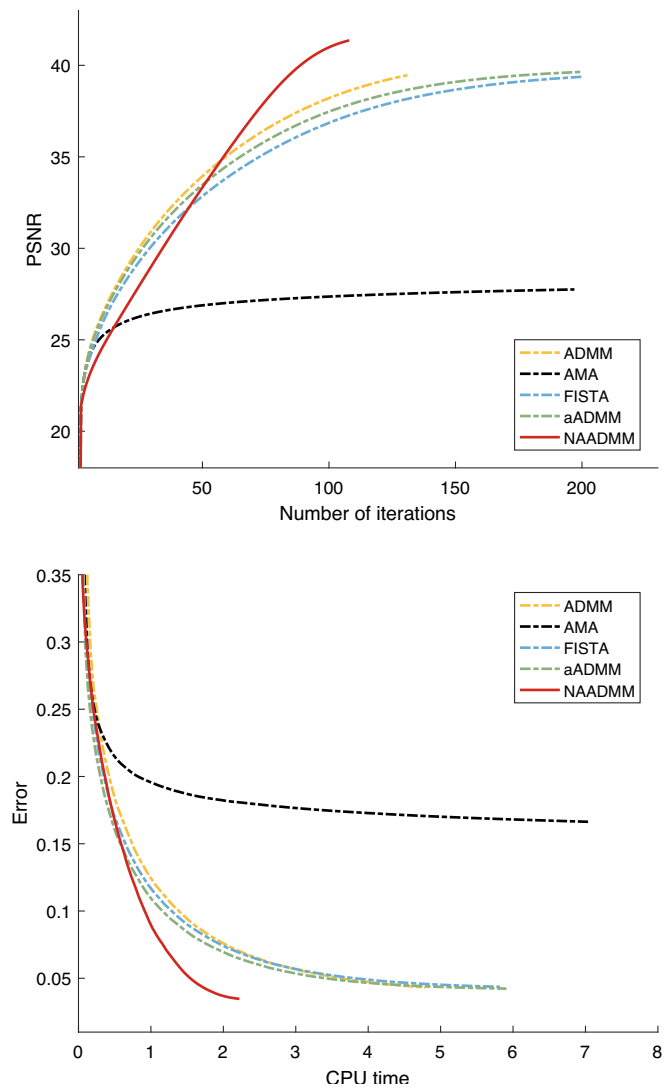

Fig. 13. Shepp Logan's MRI performance graph under (i).

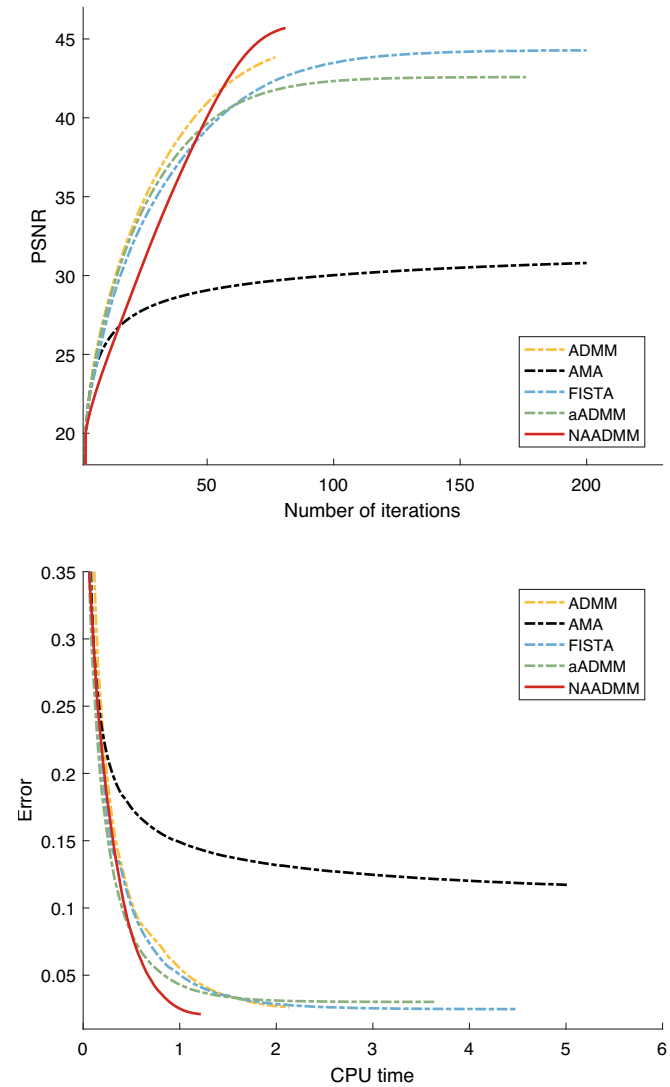

Fig. 14. Shepp Logan's MRI performance graph under (ii).

$$\nabla F(s_k) - A^T\lambda_k + \beta A^T B(\widehat{t}_k - t_k) = 0. \tag{26}$$

Similarly, substituting (11) in Algorithm 1 into the optimality condition for t_k in (10), there are

$$\nabla G(t_k) - B^T\lambda_k = 0. \tag{27}$$

By comparing (18)–(20) with (26), (27) obtained by Algorithm 1. It can be seen that the iteration point (s_k, t_k, λ_k) of Algorithm 1 satisfies the optimality condition when $b - As_k - Bt_k \to 0$, $\beta A^T B(\widehat{t}_k - t_k) \to 0$. Therefore, we define the primal and dual residuals of Algorithm 1 as follows

$$r_k = b - As_k - Bt_k, \tag{28}$$

$$d_k = A^T B(\widehat{t}_k - t_k). \tag{29}$$

The following we get the convergence about original problem (3).

Theorem 3.3. *The primal and dual residuals of Algorithm 1 satisfy the following conclusions*

$$\|r_k\|^2 \le O\left(\frac{1}{k^2}\right), \tag{30}$$

$$\|d_k\|^2 \le O\left(\frac{1}{k^2}\right), \tag{31}$$

where r_k and d_k are defined by (28), (29). Moreover Algorithm 1 produces point sequence (s_k, t_k, λ_k) that converges to the KKT point pair (s^*, t^*, λ^*) of original problem (3), which satisfies (18)–(20).

Proof : See Appendix F. □

4. Numerical experiments

Now, NAADMM algorithm 1 is used to perform MRI experiments, and then compare NAADMM algorithm with algorithms ADMM [26], aADMM [33], FISTA [18], AMA [42]. We set the experimental parameter to be $\mu = 0.01$, $\varepsilon = 0.001$, $\beta = 1.2$, two subproblems of NAADMM algorithm are solved by (15) and (16), $\eta = 0.1$. To measure the effect of the reconstructed images, we introduce the peak signal-to-noise ratio (PSNR) [52] and relative error (Error)

$$PSNR = 10log_{10}\frac{255}{MSE}dB, \quad Error = \frac{\| x_k - x_{ori} \|}{\| x_{ori} \|},$$

where x_{ori} denotes the original image, x_k denotes the recovered image, and MSE denotes the mean square error between x_{ori} and x_k. The larger the value of PSNR and the smaller the value of Error, the better the image recovery. The stopping conditions of all algorithms are chosen as follows

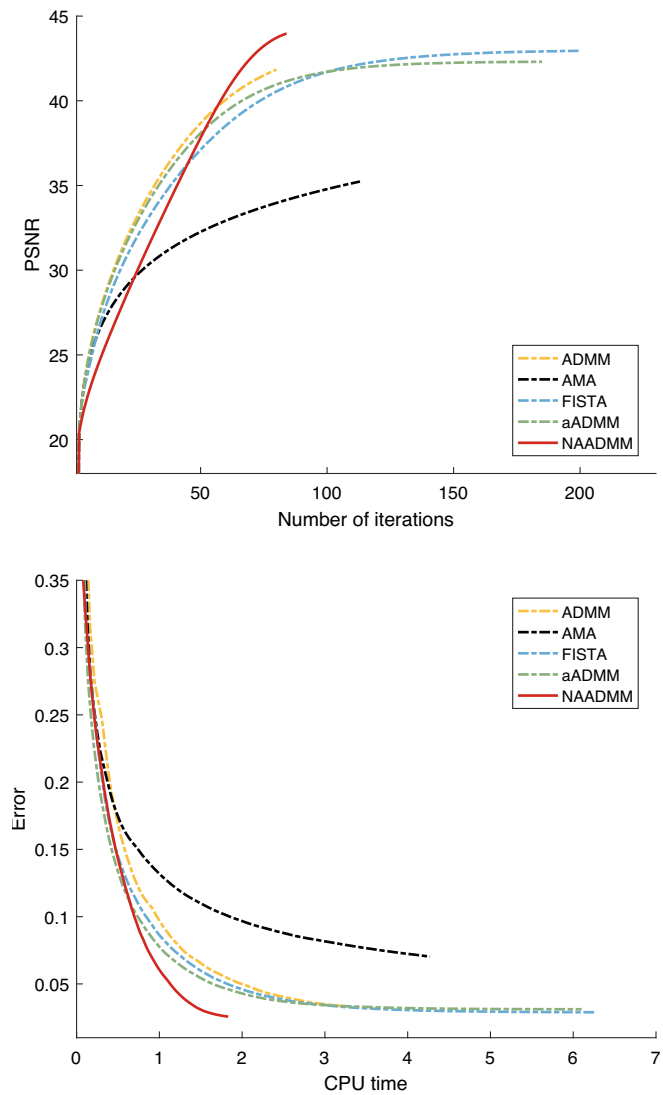

Fig. 15. Shepp Logan's MRI performance graph under (iii).

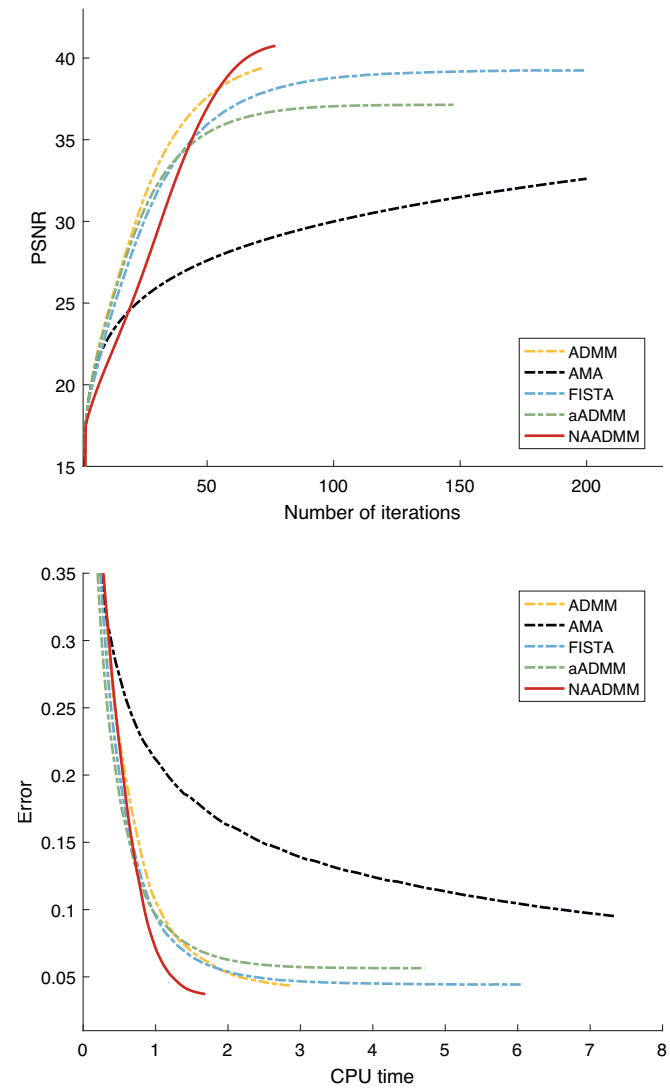

Fig. 16. Shepp Logan's MRI performance graph under (iv).

$$\frac{\|x_{k+1} - x_k\|}{\|x_k\|} < Tol \quad or \quad k < max_iter,$$

a. $Tol = 1e-4$, $max_iter = 200$,
b. $Tol = 1e-5$, $max_iter = 200$.

The two test images are Shepp Logan (256*256) [53] and Brain Angiograph (256*256), which are shown in Fig. 1.

We choose the following five templates to undersample the k-space in our experiments:

i. 70 lines of Cartesian random undersampling.
ii. 100 lines of Cartesian random undersampling.
iii. Undersampling with 30 % sampling rate and 0.1 sampling radius.
iv. Undersampling with 110 % sampling rate and 0.1 sampling radius.
v. 10 lines radially undersampled,

The five undersampling templates are shown in Fig. 2. Moreover, when the stop condition (a) is selected, Figs. 3–7 and Figs. 8–12 show the reconstruction images and error images for (i)-(v), respectively.

The numerical results of these methods under different tolerances (Tol) are given in Table 1, where CPU, Error and PSNR represent the CPU running time, relative error and the value of image recovery, respectively. Figs. 13–17 show the number of iterations-PSNR, CPU-Error performance plots for the five sampling templates, respectively. It is clear that NAADMM algorithm can obtain better reconstruction results while increasing the running speed, even when similar reconstruction results are obtained with other algorithms.

5. Conclusions

This paper is based on the TV regularisation model (1) and aims to better solve the special MRI problem (3). By combining the ideas of [41] and [43] and introducing the Nesterov acceleration technique, we propose a new accelerated ADMM, then obtain the convergence results of $O\left(1/k^2\right)$ when one objective function is strongly convex and the other is quadratically convex. In order to test the effectiveness of the NAADMM algorithm, we select five undersampled templates for testing in MRI experiment and compare it with other algorithms, the experimental data and performance graphs show that our proposed algorithm is effective and improves the quality of image recovery.

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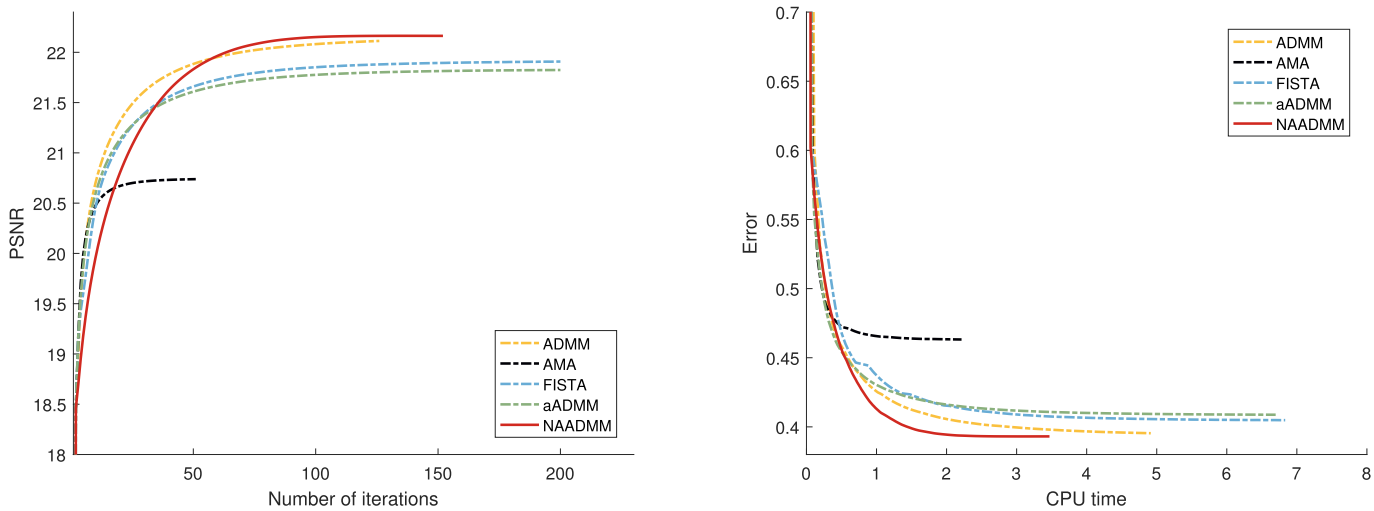

Fig. 17. Brain Angiograph's MRI performance graph under (v).

CRediT authorship contribution statement

ZhiBin Zhu: Writing – review & editing, Supervision, Methodology. **YueHong Ding:** Writing – original draft, Software, Data curation. **Ying Liu:** Data curation. **JiaQi Huang:** Data curation.

Declaration of competing interest

The authors declare no conflict of interests.

Appendix A. Appendix

A.1. Proof of Lemma 3.2

Proof: Since $F(s) = \frac{1}{2}\|Ms - y\|_2^2$, by Definition 2.4, its conjugate function is as follows

$$F^*(z) = \sup_s \left\{ z^T s - \frac{1}{2}(Ms)^T Ms + y^T Ms - \frac{1}{2} y^T y \right\}.$$

Clearly, $F^*(z)$ is the maximum value of the quadratic function $g(s) = z^T s - \frac{1}{2}(Ms)^T Ms + y^T Ms - \frac{1}{2} y^T y$ about s. So that

$$g'(s) = z^T - s^T M^T M + y^T M = 0.$$

According to Lemma 3.1

$$s = \left(M^T \mathrm{M}\right)^+ \left(z + M^T y\right).$$

However

$$F^*(z) = \frac{1}{2}\left(z + M^T y\right)^T \left(M^T \mathrm{M}\right)^+ \left(z + M^T y\right) - \frac{1}{2} y^T y.$$

Then we have

$$\nabla F^*(z) = \left(M^T \mathrm{M}\right)^+ \left(z + M^T y\right).$$

According to Theorem 2.1, and for $\forall z_1, z_2 \in R^n$, there are

$$\| \nabla F^*(z_1) - \nabla F^*(z_2) \| = \| \left(M^T \mathrm{M}\right)^+ \left(z_1 + M^T y\right) - \left(M^T \mathrm{M}\right)^+ \left(z_2 + M^T y\right) \|$$
$$\leq \| \left(M^T M\right)^+ \| \| z_1 - z_2 \|$$
$$= \frac{1}{\min\limits_{1 \leq i \leq r} \{\sigma_i\}} \| z_1 - z_2 \| .$$

In summary, by Definition 2.2, F^* has a Lipschitz continuous gradient with Lipschitz constant $L_{\nabla F^*} = \left\| \left(M^T M\right)^+ \right\|_2 = \frac{1}{\min_{1 \leq i \leq r}\{\sigma_i\}}$. According to Lemma 2.4, G^* also has a Lipschitz continuous gradient, and we compute its Lipschitz constant as follows. Since $G(t) = \mu\|t\|_1 + \frac{\varepsilon}{2}\|t\|_2^2$, by Definition 2.4, its conjugate function is as follows

$$G^*(m) = \sup_t \left\{ m^T t - \mu \| t \|_1 - \frac{\varepsilon}{2} \| t \|_2^2 \right\}$$
$$= \sup_t \left\{ \sum_{i=1}^{l} m_i^T t_i - \sum_{i=1}^{l} \mu |t_i| - \frac{\varepsilon}{2} \sum_{i=1}^{l} t_i^2 \right\}.$$

$G^*(m)$ is the maximum of the quadratic function $h(t) = \sum_{i=1}^{l} h_i(t_i) = \sum_{i=1}^{l} m_i^T t_i - \sum_{i=1}^{l} \mu|t_i| - \frac{\varepsilon}{2}\sum_{i=1}^{l} t_i^2$ about t. Since $\|\cdot\|_1$ is not smooth, we take the partial derivative of each t_i and make it to 0, we have

$$0 \in m_i - \mu\partial|t_i| - \varepsilon t_i,$$

where $\partial|t_i| = \begin{cases} 1 & t_i > 0 \\ [-1,1] & t_i = 0. \\ -1 & t_i < 0 \end{cases}$ It is clear that

a. When $t_i > 0$, $m_i - \mu - \varepsilon t_i = 0$, then $t_i = \frac{m_i - \mu}{\varepsilon}$, $h_i(t_i) = \frac{(m_i-\mu)^2}{2\varepsilon}$. At this time, $m_i > \mu$.

b. When $t_i < 0$, $m_i + \mu - \varepsilon t_i = 0$, then $t_i = \frac{m_i + \mu}{\varepsilon}$, $h_i(t_i) = \frac{{m_i}^2 - 3\mu - 2m_i\mu}{2\varepsilon}$. At this time, $m_i < -\mu$.

c. When $t_i = 0$, $\exists r \in [-1,1]$, $m_i - \mu r = 0$, then $h_i(t_i) = 0$. At this time, $-\mu \le m_i \le \mu$. Therefor

$$t_i = \left\{ \begin{array}{ll} \dfrac{m_i - \mu}{\varepsilon}, & m_i > \mu \\ \dfrac{m_i + \mu}{\varepsilon}, & m_i < -\mu \\ 0, & -\mu \le m_i \le \mu \end{array} \right\}$$

$$= \frac{max\{|m_i| - \mu, 0\} sign(m_i)}{\varepsilon}$$

$$= \frac{shrink(m_i, \mu)}{\varepsilon}.$$

By Lemma2.6. It is clear that

$$\nabla G^*(m) = \underset{t}{argmax}\{m^T t - G(t)\} = \frac{shrink(m,\mu)}{\varepsilon}.$$

We can get

$$\|\nabla G^*(m_1) - \nabla G^*(m_2)\| \le \frac{1}{\varepsilon}\|m_1 - m_2\|.$$

Similarly, according to Definition 2.2, G^* has a Lipschitz continuous gradient with Lipschitz constant $L_{\nabla G^*} = \frac{1}{\varepsilon}$.
The proof has been completed. □

Appendix B. Proof of Lemma 3.3

Proof: By the optimality condition for s_k in (9).

$$\nabla F(s_k) - A^T\widehat{\lambda}_k - \beta A^T(b - As_k - B\widehat{t}_k) = \nabla F(s_k) - A^T\overline{\lambda} = 0.$$

And so

$$A^T\overline{\lambda} = \nabla F(s_k).$$

According to Proposition 2.1, we know that

$$s_k = \nabla F^*\left(A^T\overline{\lambda}\right).$$

By the optimality condition for t_k in (10).

$$\nabla G(t_k) - B^T[\widehat{\lambda}_k + \beta(b - As_k - Bt_k)] = 0.$$

Substituting (11) into the above equation, it yields that

$$\nabla G(t_k) - B^T[\lambda_k - \beta(b - As_k - Bt_k) + \beta(b - As_k - Bt_k)] = 0.$$

Resulting in

$$\nabla G(t_k) = B^T\lambda_k.$$

Similarly, according to Proposition 2.1, we have $t_k = \nabla G^*\left(B^T\lambda_k\right)$.
In the following we prove (24) by mathematical induction, considering (13), since the initial conditions of Algorithm 1

$$\lambda_0 = \widehat{\lambda}_1, \tag{B.1}$$

$$t_0 = \widehat{t}_1 = \nabla G^*\left(B^T\lambda_0\right) = \nabla G^*\left(B^T\widehat{\lambda}_1\right). \tag{B.2}$$

1) For the case of $k = 1$, we obtain

$$\begin{aligned}
\widehat{t}_2 &= t_1 + \frac{\alpha_1 - 1}{\alpha_2}(t_1 - t_0) + \frac{\alpha_1}{\alpha_2}\left(1 - \frac{\eta}{\beta}\right)(\widehat{t}_1 - t_1) \\
&\overset{a}{=} t_1 + \frac{\alpha_1 - 1}{\alpha_2}(t_1 - t_0) + \frac{\alpha_1}{\alpha_2}\left(1 - \frac{\eta}{\beta}\right)(t_0 - t_1) \\
&\overset{b}{=} \nabla G^*\left(B^T\lambda_1\right) + \frac{\alpha_1 - 1}{\alpha_2}\left(\nabla G^*\left(B^T\lambda_1\right) - \nabla G^*\left(B^T\lambda_0\right)\right) \\
&+ \frac{\alpha_1}{\alpha_2}\left(1 - \frac{\eta}{\beta}\right)\left(\nabla G^*\left(B^T\lambda_0\right) - \nabla G^*\left(B^T\lambda_1\right)\right) \qquad \overset{c}{=} \nabla G^*\left(B^T\lambda_1\right) + \frac{\alpha_1 - 1}{\alpha_2}\left(\nabla G^*\left(B^T\lambda_1\right) - \nabla G^*\left(B^T\lambda_0\right)\right) \\
&+ \frac{\alpha_1}{\alpha_2}\left(1 - \frac{\eta}{\beta}\right)\left(\nabla G^*\left(B^T\widehat{\lambda}_1\right) - \nabla G^*\left(B^T\lambda_1\right)\right) \\
&\overset{d}{=} \nabla G^*\left(B^T\widehat{\lambda}_2\right),
\end{aligned}$$

where $=^a$ uses (B.2), $=^b$ uses (23) and (B.2), $=^c$ uses (B.1). Moreover, by Lemma 3.2, G^* is quadratic and hence ∇G^* is affine, so the above equation can be written as

$$\widehat{t}_2 = \left[1 + \frac{\alpha_1 - 1}{\alpha_2} - \frac{\alpha_1}{\alpha_2}\left(1 - \frac{\eta}{\beta}\right)\right]\nabla G^*\left(B^T\lambda_1\right) - \frac{\alpha_1 - 1}{\alpha_2}\nabla G^*\left(B^T\lambda_0\right) + \frac{\alpha_1}{\alpha_2}\left(1 - \frac{\eta}{\beta}\right)\nabla G^*\left(B^T\widehat{\lambda}_1\right).$$

Therefore

$$\left[1 + \frac{\alpha_1 - 1}{\alpha_2} - \frac{\alpha_1}{\alpha_2}\left(1 - \frac{\eta}{\beta}\right)\right] - \frac{\alpha_1 - 1}{\alpha_2} + \frac{\alpha_1}{\alpha_2}\left(1 - \frac{\eta}{\beta}\right) = 1.$$

So it is reasonable for $=^d$ to use Definition 2.5, i.e. (24) holds.

2) For the case of $k-2$, we assume that (24) holds, which makes

$$\widehat{t}_{k-1} = \nabla G^*\left(B^T\widehat{\lambda}_{k-1}\right). \tag{B.3}$$

3) For the case of $k-1$, we obtain

$$\begin{aligned}
\widehat{t}_k &= t_{k-1} + \frac{\alpha_{k-1} - 1}{\alpha_k}(t_{k-1} - t_{k-2}) + \frac{\alpha_{k-1}}{\alpha_k}\left(1 - \frac{\eta}{\beta}\right)(\widehat{t}_{k-1} - t_{k-1}) \\
&\overset{e}{=} \nabla G^*\left(B^T\lambda_{k-1}\right) + \frac{\alpha_{k-1} - 1}{\alpha_k}\left[\nabla G^*\left(B^T\lambda_{k-1}\right) - \nabla G^*\left(B^T\lambda_{k-2}\right)\right] \\
&+ \frac{\alpha_{k-1}}{\alpha_k}\left(1 - \frac{\eta}{\beta}\right)\left[\nabla G^*\left(B^T\widehat{\lambda}_{k-1}\right) - \nabla G^*\left(B^T\lambda_{k-1}\right)\right] \\
&\overset{f}{=} \nabla G^*\left(B^T\widehat{\lambda}_k\right),
\end{aligned}$$

where $=^e$ uses (23) and (B.3), $=^f$ uses Definition 2.5. In conclusion, by mathematical induction, (24) holds.

The proof has been completed. □

Appendix C. Proof of Lemma 3.4

Proof: From (11), (4) and the definition of $\overline{\lambda}$ in Lemma 3.3, which makes

$$\|\lambda_k - \overline{\lambda}\|^2 = \beta^2\|\widehat{t}_k - t_k\|^2.$$

By Lemma 2.4, Lemma 3.2, (4), (23) and (24), it follows that

$$\begin{aligned}
&\| \lambda_k - \overline{\lambda}\|^2 = \beta^2 \| \nabla G^*\left(B^T\widehat{\lambda}_k\right) - \nabla G^*\left(B^T\lambda_k\right)\|^2 \\
&\le \frac{\beta^2}{\varepsilon^2} \| \widehat{\lambda}_k - \lambda_k\|^2.
\end{aligned} \tag{C.1}$$

By (C.1), Lemma 3.2 and the convexity of F^*, for $\forall\kappa \in R^l$, we can obtain

$$\begin{aligned}
&F^*\left(A^T\kappa\right) - F^*\left(A^T\lambda_k\right) = F^*\left(A^T\kappa\right) - F^*\left(A^T\overline{\lambda}\right) + F^*\left(A^T\overline{\lambda}\right) - F^*\left(A^T\lambda_k\right) \\
&\ge \left\langle A\nabla F^*\left(A^T\overline{\lambda}\right), \kappa - \overline{\lambda}\right\rangle + F^*\left(A^T\overline{\lambda}\right) - F^*\left(A^T\lambda_k\right) \\
&\ge \left\langle A\nabla F^*\left(A^T\overline{\lambda}\right), \kappa - \overline{\lambda}\right\rangle - \left\langle A\nabla F^*\left(A^T\overline{\lambda}\right), \lambda_k - \overline{\lambda}\right\rangle \\
&- \frac{L_{\nabla F^*} \| A\|^2}{2} \| \lambda_k - \overline{\lambda}\|^2 \\
&\ge \left\langle A\nabla F^*\left(A^T\overline{\lambda}\right), \kappa - \lambda_k\right\rangle - \frac{\beta^2 L_{\nabla F^*} \| A\|^2}{2\varepsilon^2} \| \widehat{\lambda}_k - \lambda_k\|^2 \\
&\ge \left\langle A\nabla F^*\left(A^T\overline{\lambda}\right), \kappa - \lambda_k\right\rangle - \frac{1}{2\beta} \| \widehat{\lambda}_k - \lambda_k\|^2,
\end{aligned}$$

where $0 < \beta^3 \leq \frac{\varepsilon^2}{L_{\nabla F^*}\|A\|^2}$. By the convexity of G^*, there are

$$G^*\left(B^T\kappa\right) - G^*\left(B^T\lambda_k\right) \geq \left\langle \nabla G^*\left(B^T\lambda_k\right), B^T\kappa - B^T\lambda_k \right\rangle$$
$$= \left\langle B\nabla G^*\left(B^T\lambda_k\right), \kappa - \lambda_k \right\rangle.$$

Therefore

$$D(\lambda_k) - D(\kappa) = -F^*\left(A^T\lambda_k\right) - G^*\left(B^T\lambda_k\right) + \langle \lambda_k, b\rangle$$
$$+F^*\left(A^T\kappa\right) + G^*\left(B^T\kappa\right) - \langle \kappa, b\rangle$$
$$\geq \left\langle B\nabla G^*\left(B^T\lambda_k\right), \kappa - \lambda_k \right\rangle + \left\langle A\nabla F^*\left(A^T\overline{\lambda}\right), \kappa - \lambda_k \right\rangle$$
$$-\frac{1}{2\beta}\| \widehat{\lambda}_k - \lambda_k\|^2 + \langle \lambda_k - \kappa, b\rangle$$
$$= \left\langle B\nabla G^*\left(B^T\lambda_k\right) + A\nabla F^*\left(A^T\overline{\lambda}\right) - b, \kappa - \lambda_k \right\rangle - \frac{1}{2\beta}\| \widehat{\lambda}_k - \lambda_k\|^2.$$

According to Lemma 3.3, we know that

$$D(\lambda_k) - D(\kappa) \geq \langle Bt_k + As_k - b, \kappa - \lambda_k\rangle - \frac{1}{2\beta}\| \widehat{\lambda}_k - \lambda_k\|^2.$$

Substituting (11) into the above equation, we have

$$D(\lambda_k) - D(\kappa) \geq \frac{1}{\beta}\langle \widehat{\lambda}_k - \lambda_k, \kappa - \lambda_k\rangle - \frac{1}{2\beta}\| \widehat{\lambda}_k - \lambda_k\|^2$$
$$= \frac{1}{\beta}\langle \widehat{\lambda}_k - \lambda_k, \kappa - \widehat{\lambda}_k + \widehat{\lambda}_k - \lambda_k\rangle - \frac{1}{2\beta}\| \widehat{\lambda}_k - \lambda_k\|^2$$
$$= \frac{1}{\beta}\langle \widehat{\lambda}_k - \lambda_k, \kappa - \widehat{\lambda}_k\rangle + \frac{1}{2\beta}\| \widehat{\lambda}_k - \lambda_k\|^2.$$

The proof has been completed. □

Appendix D. Proof of Lemma 3.5

Proof: Considering (25), for the case of $k+1$, let $\kappa = \lambda_k$, it yields that

$$D(\lambda_{k+1}) - D(\lambda_k) \geq \frac{1}{\beta}\langle \widehat{\lambda}_{k+1} - \lambda_{k+1}, \lambda_k - \widehat{\lambda}_{k+1}\rangle + \frac{1}{2\beta}\| \widehat{\lambda}_{k+1} - \lambda_{k+1}\|^2. \tag{D.1}$$

Similarly, for the case of $k+1$, let $\kappa = \lambda^*$, we have

$$D(\lambda_{k+1}) - D(\lambda^*) \geq \frac{1}{\beta}\langle \widehat{\lambda}_{k+1} - \lambda_{k+1}, \lambda^* - \widehat{\lambda}_{k+1}\rangle + \frac{1}{2\beta}\| \widehat{\lambda}_{k+1} - \lambda_{k+1}\|^2. \tag{D.2}$$

By $(D.1) \times \alpha_{k+1}(\alpha_{k+1} - 1) + (D.2) \times \alpha_{k+1}$, there are

$$\alpha_{k+1}^2[D(\lambda_{k+1}) - D(\lambda^*)]$$
$$\geq \alpha_{k+1}^2 D(\lambda_k) - \alpha_{k+1}^2 D(\lambda^*) + \frac{\alpha_{k+1}^2}{2\beta}\| \lambda_{k+1} - \widehat{\lambda}_{k+1}\|^2$$
$$+\frac{\alpha_{k+1}^2}{\beta}\langle \widehat{\lambda}_{k+1} - \lambda_k, \lambda_{k+1} - \widehat{\lambda}_{k+1}\rangle - \alpha_{k+1}D(\lambda_k) + \alpha_{k+1}D(\lambda^*)$$
$$+\frac{\alpha_{k+1}}{\beta}\langle \widehat{\lambda}_{k+1} - \lambda^*, \lambda_{k+1} - \widehat{\lambda}_{k+1}\rangle - \frac{\alpha_{k+1}}{\beta}\langle \widehat{\lambda}_{k+1} - \lambda_k, \lambda_{k+1} - \widehat{\lambda}_{k+1}\rangle$$
$$= \frac{\alpha_{k+1}}{\beta}\langle (\widehat{\lambda}_{k+1} - \lambda_k)(\alpha_{k+1} - 1) + \widehat{\lambda}_{k+1} - \lambda^*, \lambda_{k+1} - \widehat{\lambda}_{k+1}\rangle$$
$$+\frac{\alpha_{k+1}^2}{2\beta}\| \lambda_{k+1} - \widehat{\lambda}_{k+1}\|^2 + \alpha_{k+1}(\alpha_{k+1} - 1)[D(\lambda_k) - D(\lambda^*)].$$

Using $\langle x, y\rangle = \frac{1}{2\eta}[\| \eta x + y\|^2 - \| \eta x\|^2 - \| y\|^2]$ for the inner product in the above equation, which makes

$$
\begin{aligned}
&\alpha_{k+1}^2[D(\lambda_{k+1})-D(\lambda^*)]\\
&\geq \alpha_{k+1}(\alpha_{k+1}-1)[D(\lambda_k)-D(\lambda^*)]\\
&+\frac{1}{2\eta}\|\frac{\eta\alpha_{k+1}}{\beta}(\lambda_{k+1}-\widehat{\lambda}_{k+1})+(\widehat{\lambda}_{k+1}-\lambda_k)(\alpha_{k+1}-1)+\widehat{\lambda}_{k+1}-\lambda^*\|^2\\
&-\frac{1}{2\eta}\|(\widehat{\lambda}_{k+1}-\lambda_k)(\alpha_{k+1}-1)+\widehat{\lambda}_{k+1}-\lambda^*\|^2\\
&+\frac{\alpha_{k+1}^2}{2\beta}\|\lambda_{k+1}-\widehat{\lambda}_{k+1}\|^2-\frac{1}{2\eta}\|\eta\cdot\frac{\alpha_{k+1}}{\beta}(\lambda_{k+1}-\widehat{\lambda}_{k+1})\|^2\\
&\geq \alpha_{k+1}(\alpha_{k+1}-1)[D(\lambda_k)-D(\lambda^*)]\\
&+\frac{1}{2\eta}\|\frac{\eta\alpha_{k+1}}{\beta}(\lambda_{k+1}-\widehat{\lambda}_{k+1})+(\widehat{\lambda}_{k+1}-\lambda_k)(\alpha_{k+1}-1)+\widehat{\lambda}_{k+1}-\lambda^*\|^2\\
&-\frac{1}{2\eta}\|(\widehat{\lambda}_{k+1}-\lambda_k)(\alpha_{k+1}-1)+\widehat{\lambda}_{k+1}-\lambda^*\|^2+\frac{\alpha_{k+1}^2}{2\beta}\left(1-\frac{\eta}{\beta}\right)\|\lambda_{k+1}-\widehat{\lambda}_{k+1}\|^2.
\end{aligned}
$$

Since $0<\eta<\beta$, we obtain

$$
\begin{aligned}
&\alpha_{k+1}^2[D(\lambda_{k+1})-D(\lambda^*)]\\
&\geq \alpha_{k+1}(\alpha_{k+1}-1)[D(\lambda_k)-D(\lambda^*)]\\
&+\frac{1}{2\eta}\|\frac{\eta\alpha_{k+1}}{\beta}(\lambda_{k+1}-\widehat{\lambda}_{k+1})+(\widehat{\lambda}_{k+1}-\lambda_k)(\alpha_{k+1}-1)+\widehat{\lambda}_{k+1}-\lambda^*\|^2\\
&-\frac{1}{2\eta}\|(\widehat{\lambda}_{k+1}-\lambda_k)(\alpha_{k+1}-1)+\widehat{\lambda}_{k+1}-\lambda^*\|^2.
\end{aligned}
\tag{D.3}
$$

Since the following equation holds

$$
m_k=\frac{\eta\alpha_k}{\beta}(\lambda_k-\widehat{\lambda}_k)+(\widehat{\lambda}_k-\lambda_{k-1})(\alpha_k-1)+\widehat{\lambda}_k-\lambda^*. \tag{D.4}
$$

Substituting (D.4) and (14) into (D.3), it follows that

$$
\alpha_{k+1}^2[D(\lambda_{k+1})-D(\lambda^*)]\geq \alpha_{k+1}(\alpha_{k+1}-1)[D(\lambda_k)-D(\lambda^*)]+\frac{1}{2\eta}\|m_{k+1}\|^2-\frac{1}{2\eta}\|m_k\|^2.
$$

By (12) in Algorithm 1, $\alpha_{k+1}(\alpha_{k+1}-1)=\alpha_k^2$ holds, so that

$$
\|m_{k+1}\|^2+2\eta\alpha_{k+1}^2[D(\lambda^*)-D(\lambda_{k+1})]\leq\|m_k\|^2+2\eta\alpha_k^2[D(\lambda^*)-D(\lambda_k)].
$$

The proof has been completed. □

Appendix E. Proof of Theorem 3.2

Proof: In (D.2), let $k=0$ then we have

$$
\begin{aligned}
&D(\lambda_1)-D(\lambda^*)\geq\frac{1}{2\beta}\|\lambda_1-\widehat{\lambda}_1\|^2+\frac{1}{\beta}\langle\widehat{\lambda}_1-\lambda^*,\lambda_1-\widehat{\lambda}_1\rangle\\
&=\frac{1}{2\beta}\left[\|\lambda_1-\lambda^*\|^2-\|\widehat{\lambda}_1-\lambda^*\|^2\right].
\end{aligned}
$$

By Lemma 3.5, the sequence $\left\{\|m_k\|^2+2\eta\alpha_k^2[D(\lambda^*)-D(\lambda_k)]\right\}$ is monotonically decreasing, there are

$$
\begin{aligned}
&\|m_{k+1}\|^2+2\eta\alpha_{k+1}^2[D(\lambda^*)-D(\lambda_{k+1})]\leq\|m_k\|^2+2\eta\alpha_k^2[D(\lambda^*)-D(\lambda_k)]\\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\leq\cdots\\
&\leq\|m_1\|^2+2\eta\alpha_1^2[D(\lambda^*)-D(\lambda_1)].
\end{aligned}
$$

By (D.4), let $k=1$, it yields that

$$
\begin{aligned}
&2\eta\alpha_{k+1}^2[D(\lambda^*)-D(\lambda_{k+1})]\\
&\leq\|m_1\|^2+2\eta\alpha_1^2[D(\lambda^*)-D(\lambda_1)]\\
&=\|\frac{\eta}{\beta}(\lambda_1-\widehat{\lambda}_1)+\widehat{\lambda}_1-\lambda^*\|^2+2\eta[D(\lambda^*)-D(\lambda_1)]\\
&\leq\frac{2\eta^2}{\beta^2}\|\lambda_1-\widehat{\lambda}_1\|^2+2\|\widehat{\lambda}_1-\lambda^*\|^2+\frac{\eta}{\beta}\left(\|\widehat{\lambda}_1-\lambda^*\|^2-\|\lambda_1-\lambda^*\|^2\right)\\
&\leq\frac{2\eta^2}{\beta^2}\|\lambda_1-\widehat{\lambda}_1\|^2+2\|\widehat{\lambda}_1-\lambda^*\|^2+\frac{\eta}{\beta}\|\widehat{\lambda}_1-\lambda^*\|^2\\
&=\frac{2\eta^2}{\beta^2}\|\lambda_1-\widehat{\lambda}_1\|^2+\frac{\eta+2\beta}{\beta}\|\widehat{\lambda}_1-\lambda^*\|^2.
\end{aligned}
$$

Resulting in

$$D(\lambda^*) - D(\lambda_{k+1}) \leq \frac{\frac{2\eta}{\beta^2}\|\lambda_1 - \widehat{\lambda}_1\|^2 + \frac{\eta+2\beta}{\eta\beta}\|\widehat{\lambda}_1 - \lambda^*\|^2}{2\alpha_{k+1}^2}.$$

According to (12) in Algorithm 1, which makes

$$\alpha_{k+1} > \frac{1}{2} + \alpha_k > \cdots > \frac{1}{2} + \cdots + \frac{1}{2} + \alpha_{k-(k-1)} = \frac{k+2}{2}.$$

So there are

$$D(\lambda^*) - D(\lambda_{k+1}) \leq \frac{\frac{4\eta}{\beta^2}\|\lambda_1 - \widehat{\lambda}_1\|^2 + \frac{2\eta+4\beta}{\eta\beta}\|\widehat{\lambda}_1 - \lambda^*\|^2}{(k+2)^2}.$$

The proof has been completed. □

Appendix F. Proof of Theorem 3.3

Proof: Since F is quadratic convex and G is strongly convex, then $-D$ is strongly convex, then we assume it is μ_D- strongly convex, and by Lemma 2.3, we can get

$$D(\lambda^*) - D(\lambda_k) \geq \frac{\mu_D}{2}\|\lambda^* - \lambda_k\|^2.$$

Combining with Theorem 3.2, it yields that

$$\frac{\mu_D}{2} \parallel \lambda_k - \lambda^* \|^2 \leq [D(\lambda^*) - D(\lambda_k)]$$
$$\leq \frac{\frac{4\eta}{\beta^2} \parallel \lambda_1 - \widehat{\lambda}_1 \|^2 + \frac{2\eta+4\beta}{\eta\beta} \parallel \widehat{\lambda}_1 - \lambda^* \|^2}{(k+1)^2}.$$

Therefore

$$\|\lambda_k - \lambda^*\|^2 \leq O\left(\frac{1}{k^2}\right). \tag{F.1}$$

Resulting in

$$\|\lambda_{k+1} - \lambda_k\|^2 \leq \|\lambda_{k+1} - \lambda^*\|^2 + \|\lambda^* - \lambda_k\|^2 \leq O\left(\frac{1}{k^2}\right).$$

By Lemma 3.2, D is also a function with a Lipchitz continuous gradient $L_{\nabla D}$, and by Lemma 2.2, for $\forall k \geq 0$, so that

$$D(\widehat{\lambda}_k) \geq D(\lambda^*) + \langle \nabla D(\lambda^*), \widehat{\lambda}_k - \lambda^* \rangle - \frac{L_{\nabla D}}{2}\|\widehat{\lambda}_k - \lambda^*\|^2.$$

Combining with Theorem 3.1, this gives

$$D(\widehat{\lambda}_k) \geq D(\lambda^*) - \frac{L_{\nabla D}}{2}\|\widehat{\lambda}_k - \lambda^*\|^2.$$

For the above equation, taking $k+1$, we have

$$D(\lambda_{k+1}) - D(\widehat{\lambda}_{k+1})$$
$$\leq D(\lambda_{k+1}) - D(\lambda^*) + \frac{L_{\nabla D}}{2}\|\widehat{\lambda}_{k+1} - \lambda^*\|^2.$$

Since $D(\lambda_{k+1}) \leq D(\lambda^*)$, it follows that

$$D(\lambda_{k+1}) - D(\widehat{\lambda}_{k+1}) \leq \frac{L_{\nabla D}}{2}\|\widehat{\lambda}_{k+1} - \lambda^*\|^2.$$

Substituting (14) into the above equation, which makes

$$
\begin{aligned}
&D(\lambda_{k+1}) - D(\widehat{\lambda}_{k+1}) \\
&\le \frac{L_{\nabla D}}{2} \| \widehat{\lambda}_{k+1} - \lambda^* \|^2 \\
&= \frac{L_{\nabla D}}{2} \| \lambda_k - \lambda^* + \frac{(\alpha_k - 1)}{\alpha_{k+1}}(\lambda_k - \lambda_{k-1}) + \frac{\alpha_k}{\alpha_{k+1}}\left(1 - \frac{\eta}{\beta}\right)(\widehat{\lambda}_k - \lambda_k)\|^2 \\
&\le \frac{3L_{\nabla D}}{2} \| \lambda_k - \lambda^* \|^2 + \frac{3L_{\nabla D}}{2} \frac{(\alpha_k - 1)^2}{\alpha_{k+1}^2} \| \lambda_k - \lambda_{k-1} \|^2 \\
&+ \frac{3L_{\nabla D}}{2} \frac{\alpha_k^2}{\alpha_{k+1}^2}\left(1 - \frac{\eta}{\beta}\right)^2 \| \widehat{\lambda}_k - \lambda_k \|^2 \\
&\le O\left(\frac{1}{k^2}\right) + \frac{3L_{\nabla D}}{2} \frac{\alpha_k^2}{\alpha_{k+1}^2}\left(1 - \frac{\eta}{\beta}\right)^2 \| \widehat{\lambda}_k - \lambda_k \|^2 \\
&\le O\left(\frac{1}{k^2}\right) + \frac{3\alpha_k^2 L_{\nabla D}}{\alpha_{k+1}^2}\left(1 - \frac{\eta}{\beta}\right)^2 \| \widehat{\lambda}_k - \lambda^* \|^2 + \frac{3\alpha_k^2 L_{\nabla D}}{\alpha_{k+1}^2}\left(1 - \frac{\eta}{\beta}\right)^2 \| \lambda^* - \lambda_k \|^2 \\
&\le O\left(\frac{1}{k^2}\right) + \frac{3\alpha_k^2 L_{\nabla D}}{\alpha_{k+1}^2}\left(1 - \frac{\eta}{\beta}\right)^2 \| \widehat{\lambda}_k - \lambda^* \|^2 \\
&= O\left(\frac{1}{k^2}\right) + \frac{L_{\nabla D}}{2} \frac{6\alpha_k^2}{\alpha_{k+1}^2}\left(1 - \frac{\eta}{\beta}\right)^2 \| \widehat{\lambda}_k - \lambda^* \|^2 \\
&\le \cdots \cdots \le O\left(\frac{1}{k^2}\right) + \frac{L_{\nabla D}}{2} \frac{6^k \alpha_1^2}{\alpha_{k+1}^2}\left(1 - \frac{\eta}{\beta}\right)^{2^k} \| \widehat{\lambda}_1 - \lambda^* \|^2 \\
&= O\left(\frac{1}{k^2}\right) + \frac{L_{\nabla D}}{2} \frac{1}{\alpha_{k+1}^2}\left[6\left(1 - \frac{\eta}{\beta}\right)^2\right]^k \| \widehat{\lambda}_1 - \lambda^* \|^2 \\
&\overset{f}{\le} O\left(\frac{1}{k^2}\right),
\end{aligned}
$$

where $\overset{f}{\le}$ uses the fact $1 - \sqrt{\frac{1}{6}} < \frac{\eta}{\beta} < 1$.

Considering (25), for the case of $k+1$, let $\kappa = \widehat{\lambda}_{k+1}$, and combining with the above equation, there are

$$O\left(\frac{1}{k^2}\right) \ge D(\lambda_{k+1}) - D(\widehat{\lambda}_{k+1}) \ge \frac{1}{2\beta}\|\widehat{\lambda}_{k+1} - \lambda_{k+1}\|^2. \tag{F.2}$$

This gives us the following bound

$$\|\widehat{\lambda}_{k+1} - \lambda_{k+1}\|^2 \le O\left(\frac{1}{k^2}\right). \tag{F.3}$$

Substituting (11) into (F.2), it yields that

$$
\begin{aligned}
&D(\lambda_{k+1}) - D(\widehat{\lambda}_{k+1}) \ge \frac{1}{2\beta} \| \beta(b - As_{k+1} - Bt_{k+1}) \|^2 \\
&= \frac{\beta}{2} \| r_{k+1} \|^2.
\end{aligned}
$$

So there are

$$\|r_k\|^2 \le O\left(\frac{1}{k^2}\right). \tag{F.4}$$

For the dual residuals, according to Lemma 3.2, Lemma 3.3, (4) and (F.3), there are

$$
\begin{aligned}
&\| d_k \|^2 = \| A^T B(\widehat{t}_k - t_k) \|^2 \\
&= \| A^T B\left[\nabla G^*\left(B^T \widehat{\lambda}_k\right) - \nabla G^*\left(B^T \lambda_k\right)\right] \|^2 \\
&\le \frac{\| A \|^2}{\varepsilon^2} \| \widehat{\lambda}_k - \lambda_k \|^2 \\
&\le O\left(\frac{1}{k^2}\right).
\end{aligned}
$$

By (F.1), we know that

$$\lambda_k \to \lambda^*, for\ k \to \infty. \tag{F.5}$$

Considering (F.1) and (F.3), we can get

$$\| \widehat{\lambda}_k - \lambda^* \|^2 = \| \widehat{\lambda}_k - \lambda_k + \lambda_k - \lambda^* \|^2$$
$$\leq 2 \| \widehat{\lambda}_k - \lambda_k \|^2 + 2 \| \lambda_k - \lambda^* \|^2$$
$$\leq O\left(\frac{1}{k^2}\right).$$

So there are

$$\widehat{\lambda}_k \to \lambda^*, for\ k \to \infty. \tag{F.6}$$

By the definition of $\overline{\lambda}$ in Lemma 3.3, which satisfy that

$$A^T\overline{\lambda} = A^T\widehat{\lambda}_k + \beta A^T(b - As_k - Bt_k + Bt_k - B\widehat{t}_k).$$

Considering (30), (31) and (F.6), it is clear that

$$A^T\overline{\lambda} \to A^T\lambda^*.$$

Combining (22) and the above equation, so that

$$s_k \to s^*, for\ k \to \infty.$$

In addition, according to (23) and (F.5), it can be obtained that

$$t_k \to t^*, for\ k \to \infty.$$

In summary, this gives

$$(s_k, t_k, \lambda_k) \to (s^*, t^*, \lambda^*), for\ k \to \infty.$$

The proof has been completed. □

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